EXAM PAPERS PRACTICE

N3 Modelling with probability distributions A level only

Name:

Class:

Date:

Time: **118 minutes**

Marks: **99 marks**

Comments:

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Q1.

Rahul works for a bank. He can travel to the branch where he works by car, bus or bicycle. He does not wish to arrive too early as he cannot enter the building before it is unlocked at 8.30 am. He does not wish to arrive after his starting time, which is 9.00 am.

(a) When he travels by bus his journey time may be modelled by a normally distributed random variable with mean 40 minutes and standard deviation 10 minutes. Given that he leaves home at 8.00 am and travels by bus, find the probability that he will arrive at the branch:

(i) before 9.00 am;

(ii) between 8.30 am and 9.00 am.

(5)

(b) When he travels by bicycle his journey time may be modelled by a normally distributed random variable with mean 35 minutes and standard deviation 2 minutes. What time should he leave home when he travels by bicycle if he wishes to have the probability of 0.99 of arriving before 9.00 am?

(5)

(c) When he travels by car his journey time may be modelled by a normally distributed random variable with mean 28 minutes and standard deviation 8 minutes. Explain briefly why, when he travels by car, he should leave home at 8.17 am in order to maximise his chance of arriving between 8.30 am and 9.00 am.

(2)

(d) Give an advantage of:

(i) travelling by car compared to bus or bicycle;

(1)

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(ii) travelling by bicycle compared to bus or car.

(1)

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Your answers should be based only on the data given in the question and not on general considerations such as exercise, environmental consequences and comfort.

(Total 14 marks)

Q2.

Each house in a local authority area is supplied with a recycling box for discarded bottles. The council arranges a fortnightly collection of the boxes from outside the houses.

In a street containing 30 houses the number of boxes placed outside for a particular collection is a random variable X. The values of X for 9 consecutive collections were:

15 12 10 11 12 14 11 10 13

(a) Calculate the mean and standard deviation of the given data.

(2)

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(b) Use the data to estimate p, the proportion of houses that a box will be placed outside for a particular collection.

(1)

(c) Suggest the name of a distribution which might provide a suitable model for the random variable X.

(1)

(d) Use the distribution you have suggested in part (c) and the value of p you have estimated in part (b) to find:

(i) the probability that X exceeds 15;

(3)

(ii) the mean and standard deviation of X.

(3)

(e) By comparing your answers to part (a) with your answers to part (d)(ii), explain whether the distribution you have suggested in part (c) does provide a suitable model for X.

(2)

(f) Regardless of your calculations and of your answer to part (e), give a reason why the distribution you have suggested in part (c) may not provide a suitable model for X.

(1)
(Total 13 marks)

Q3.

A gas supplier maintains a team of engineers who are available to deal with leaks reported by customers. Most reported leaks can be dealt with quickly but some require a long time. The time (excluding travelling time) taken to deal with reported leaks is found to have a mean of 65 minutes and a standard deviation of 60 minutes.

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(a) Assuming that the times may be modelled by a normal distribution, estimate the probability that:

(i) it will take more than 185 minutes to deal with a reported leak;

(3)

(ii) it will take between 50 minutes and 125 minutes to deal with a reported leak;

(4)

(b) A statistician, consulted by the gas supplier, stated that, as the times had a mean of 65 minutes and a standard deviation of 60 minutes, the normal distribution would not provide an adequate model.

Explain the reason for the statistician's statement.

(2)
(Total 9 marks)

Q4.

Charlotte is a dentist. The time that each patient, attending for an appointment, spends in her surgery may be modelled by a normal distribution with mean 23 minutes and standard deviation 4 minutes.

(a) Find the probability that a patient attending for an appointment will spend:

(i) less than 28 minutes in her surgery;

(3)

(ii) between 20 and 30 minutes in her surgery.

(4)

(b) Find the length of time spent in her surgery which will be exceeded by 1% of patients.

(4)

(c) Calculate limits, symmetrical about the mean, within which 90% of times spent by patients in her surgery will lie.

(4)

(Total 15 marks)

Q5.

The number of miles that Anita's motorbike will travel on one gallon of petrol may be modelled by a normal distribution with mean 135 and standard deviation 12.

(a) Given that Anita starts a journey with one gallon of petrol in her motorbike's tank, find the probability that, without refuelling, she can travel:

(i) more than 111 miles;

(3)

(ii) between 141 and 150 miles.

(4)

(b) Find the longest journey that Anita can undertake, if she is to have a probability of at least 0.9 of completing it on one gallon of petrol.

(4)

(Total 11 marks)

Q6.

Siballi is a student who travels to and from university by bus. He believes that the probability of having to wait more than 6 minutes to catch a bus is 0.4 and is independent of the time of day and direction of travel.

(a) Assuming that Siballi's beliefs are correct, find the probability that, during a particular week when he catches 10 buses, the number of times he has to wait more than 6 minutes is:

(i) 3 or fewer;

(3)

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(ii) more than 4.

(2)

(b) Assuming that Siballi's beliefs are correct, calculate values for the mean and standard deviation of the number of times he has to wait more than six minutes to catch a bus in a week when he catches 10 buses.

(3)

(c) During a thirteen-week period, the numbers of times (out of 10) he had to wait more than six minutes to catch a bus were as follows:

4 8 8 9 3 2 2 7 0 1 5 2 0

(i) Calculate the mean and standard deviation of these data.

(2)

(ii) State, giving reasons, whether your answers to part (c)(i) support Siballi's beliefs that the probability of having to wait more than 6 minutes to catch a bus is 0.4 and is independent of the time of day and direction of travel.

(3)

(Total 13 marks)

Q7.

A meter is used to monitor the amount of dust in a factory atmosphere. A large number of readings was taken when the atmosphere was satisfactory and it was found that these readings could be modelled by a normal distribution with mean 266 and standard deviation 32.

(a) Find the probability that, when the atmosphere is satisfactory, the meter will give a reading between 250 and 300.

(5)

(b) The factory Health and Safety Committee recommends that action be taken to purify the atmosphere if the machine gives a reading greater than k. The value of k is chosen so that the probability of the meter giving a reading greater than k, when the atmosphere is satisfactory, is 0.04.

Find the value of k.

(4)

(Total 9 marks)

Q8.

A study showed that the time, T in minutes, spent by a customer between entering and leaving Fely's department store has a mean of 20 with a standard deviation of 6.

Assume that T may be modelled by a normal distribution.

(a) Find the value of T exceeded by 20% of customers.

(4)

(b) (i) Write down the standard deviation of the mean time spent in Fely's store by a random sample of 90 customers.

(1)

(ii) Find the probability that this mean time will exceed 21 minutes.

(4)

(Total 9 marks)

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